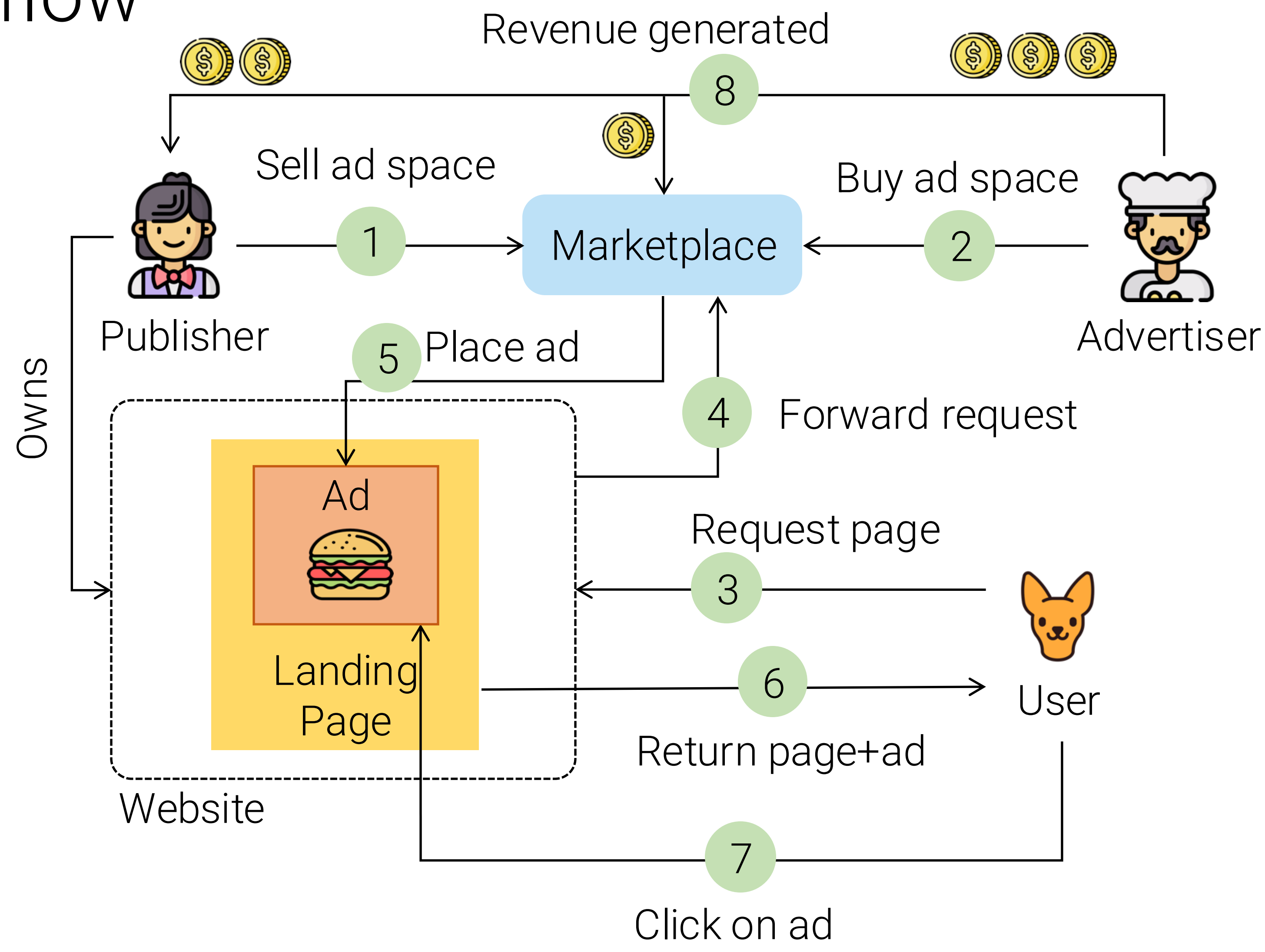
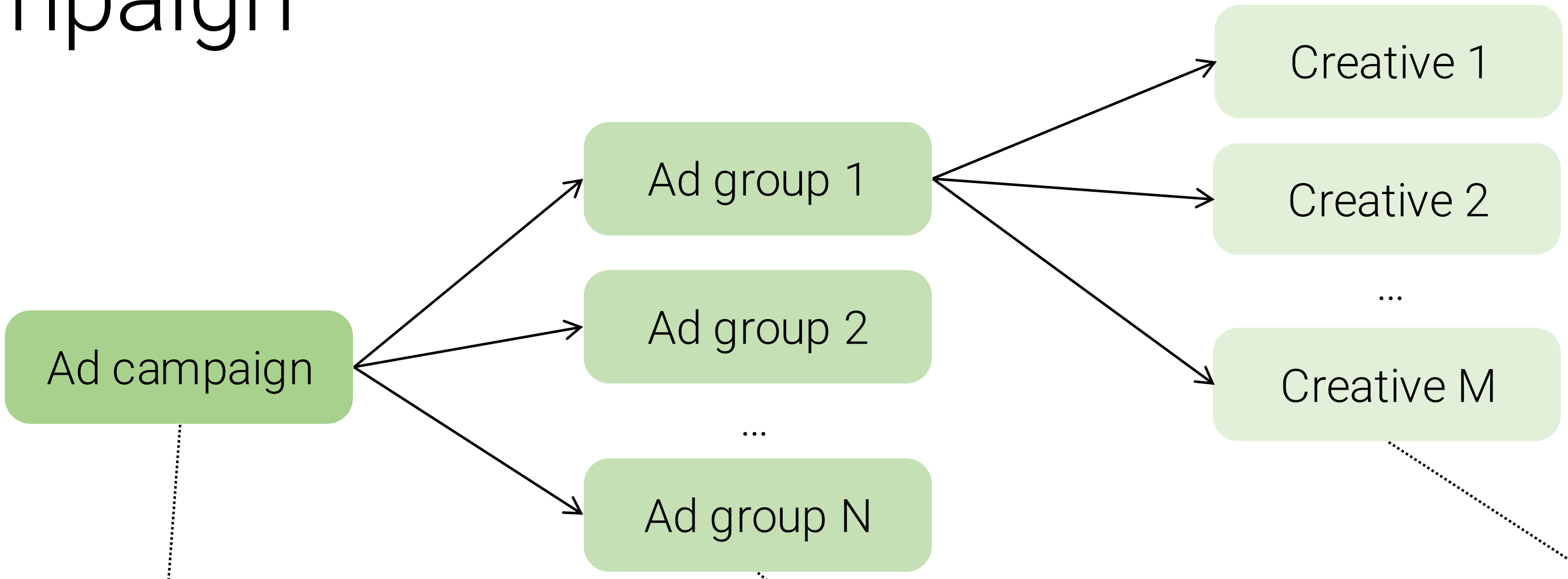
Actors & ad serving workflow

- Publisher
  - Owns of a webpage, wants to get money by showing ads
- Advertiser
  - Interested in advertising its products to users, willing to pay money for online ad space
- Ad exchange (manager)
  - A broker between publishers and advertisers, dynamically matches users to ads to maximize revenue (for a fee)
- User
  - Browses content and prefers relevant ads. Its views and clicks will generate revenue



- Advertiser might coincide with publisher

Ad campaign



The "contract" between the advertiser and the marketplace

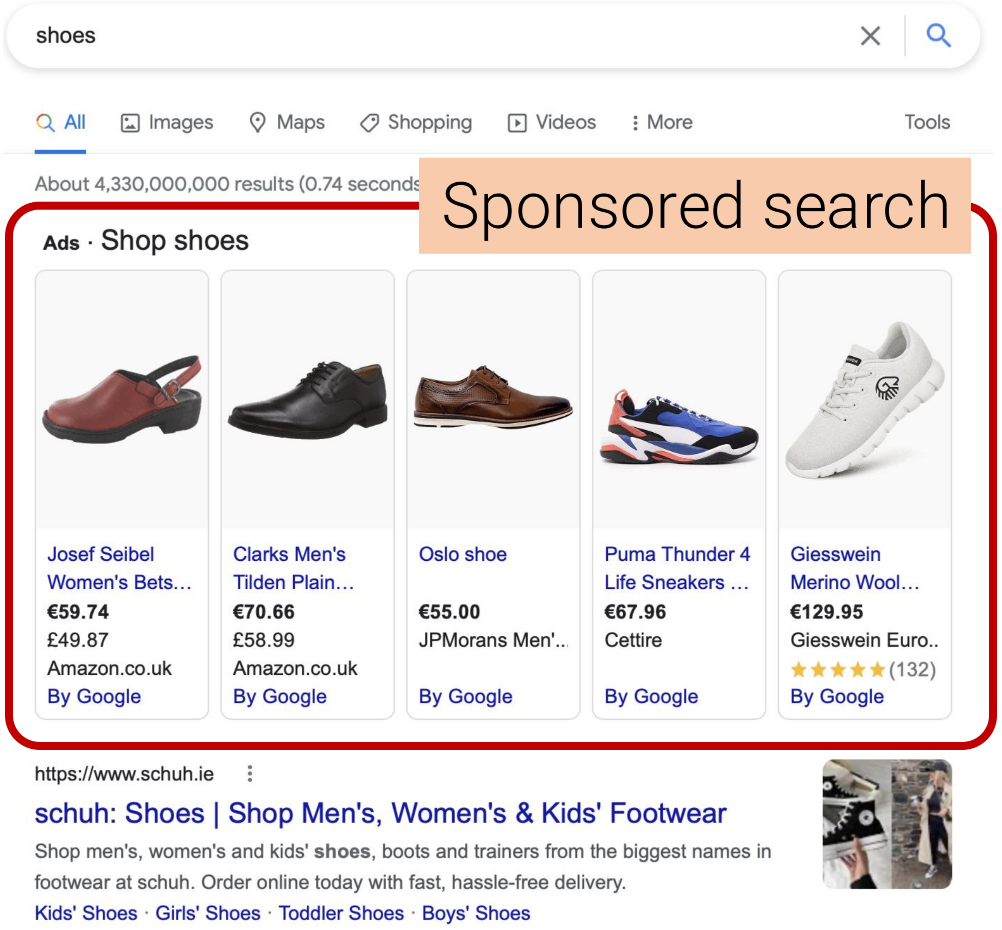
- Budget
- Timeframe

A set of ads that share a common marketing strategy

- Audience targeting
- Bid price

The actual ad

Categories of ads



Video Ads

- Sponsored search
  - Query-matching ads
- Display advertising
  - Banners on different sections of a website
- Native ads
  - Ads take the form of the platform's content

Advertising effectiveness

$$\underbrace{\text{Exposure} \rightarrow \text{Attention}}_{\text{Selection}} \rightarrow \underbrace{\text{Comprehension} \rightarrow \text{Acceptance}}_{\text{Interpretation}} \rightarrow \underbrace{\text{Retention} \rightarrow \text{Decision}}_{\text{Attitude}}$$

- **Exposure**: the ad is displayed to the user
- **Attention**: the user becomes aware of the ad
- **Comprehension**: the user understands what the ad is about
- **Acceptance**: the user finds the ad relevant to some degree
- **Retention**: the user remembers the (content of the) ad long-term
- **Decision**: a “conversion” happens (e.g., user buys or subscribes)

Billing models

- **Cost Per Time (CPT)** Exposure
  - Price billed in advance for a given advertising space, for a limited time
  - E.g., Pay 100$ to display ad X on the top banner of `mynewswebsite.com` for 1 day
- **Cost Per Mille (CPM)** or **Cost Per Impression (CPI)** Exposure
  - Price billed for every (thousand) impression(s) of the advertisement
  - E.g., Pay 1$ for every 1000 users who see ad on `mynewswebsite.com` page
- **Cost Per Sale (CPS)** or **Cost Per Action (CPA)** or **Pay Per Lead (PPL)** Decision
  - Price billed for every completed transaction or action triggered by an advertisement
  - E.g., Pay 5$ for every pair of shoes on `myshoestore.com` after clicking on the ad
- **Cost Per Click (CPC)** or **Pay Per Click (PPC)** Acceptance
  - Price billed for every click on the ad (the basis of the performance-based advertising)

Ad performance

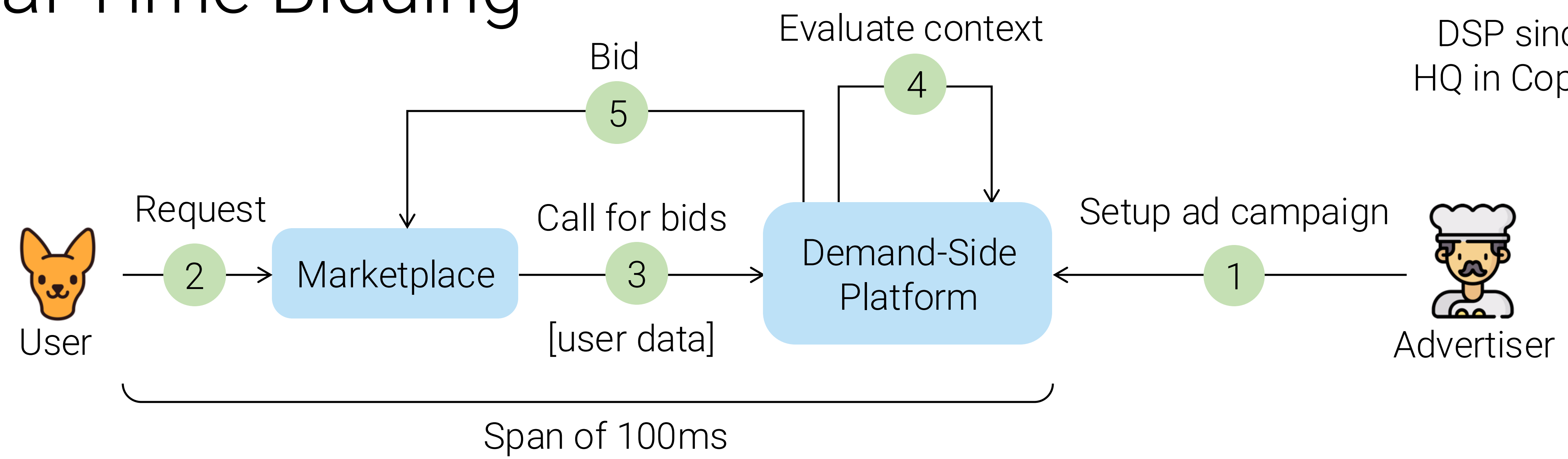
Usually measured at the exposure/acceptance level

- Click Through Rate (CTR)
  - Ratio of clicks to impressions
- View Through Rate (VTR)
  - Ratio of media item views per impression

Billing agreements

- Agreement-based
  - Marketplace, advertiser, and publisher agree in advance on a price for a given service
- Auction-based
  - Multiple advertisers bid for a publisher’s resource (a space, a query), the marketplace

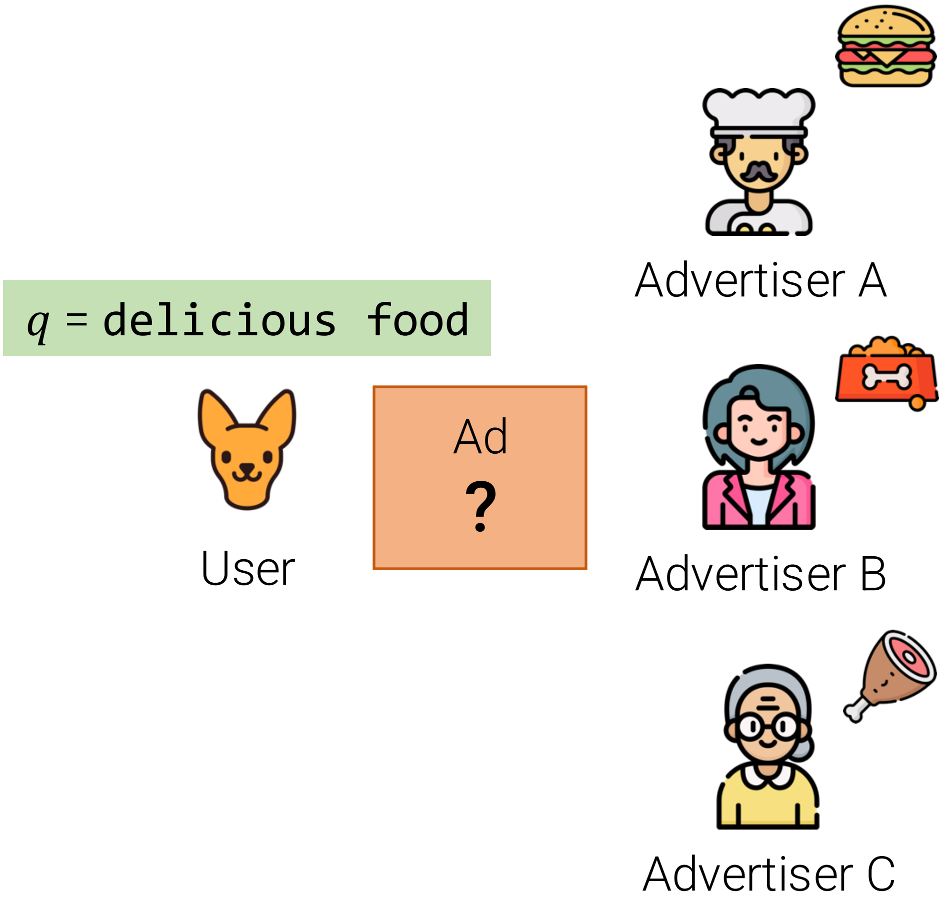
Real Time Bidding



- Bids can be done in real-time, for each user request
- Marketplaces issue call for bids for a specific (user,context)
- Demand-Side Platforms (DSPs) manage ad campaigns
  - Receive the call for bids
  - Decide whether to bid and how much based on context and performance
- The whole cycle happens in real time (100ms)

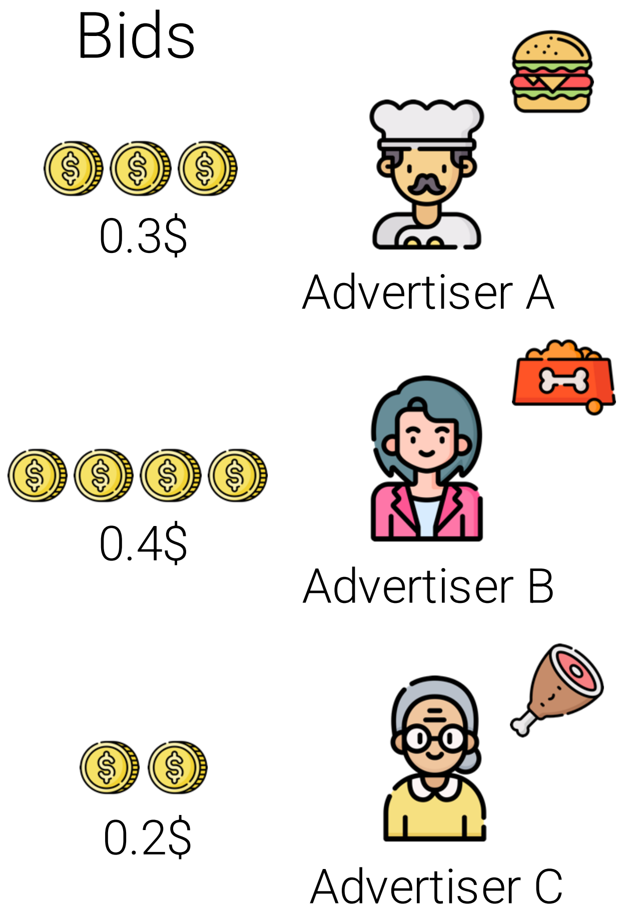
Performance-based advertising

- Advertisers bid on a set of queries Q
  - Broad matching (fuzzy similarity, wildcard queries) is often used
- A query $q \in Q$ arrives
- Need to select an ad a to match query q such that revenue is maximized
- Queries arrive live and corresponding ads must be served in a streaming fashion
- An online algorithm is needed



Simple: rank by bid!

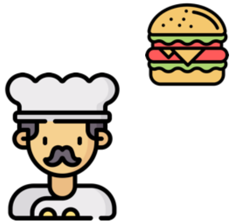
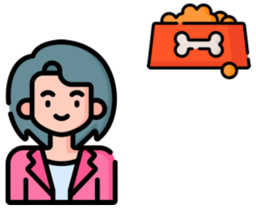
- Not really… as ads with high bid might yield no revenue if nobody clicks on them
- Sorting ads by bid might not be optimal, if users end up not clicking on the ad



Rank by expected revenue

q = `delicious food`

- Idea: sort directly by the objective measure (expected revenue)
  - Estimate CTR of query,ad pair
  - Sort by bid x CTR

Expected revenue Bid x CTR(ad, q)	Estimated CTR(ad, q)	Bids	
0.02*0.3 = 0.006$	2%	0.3$	 Advertiser A
0.01*0.4 = 0.004$	1%	0.4$	 Advertiser B
0.05*0.2 = $\boxed{0.01\$}$	5%	0.2$	 Advertiser C

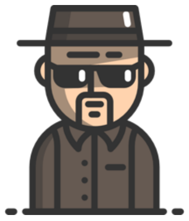
Highest expected revenue

The relevance filter

- A simple bid x CTR model is prone to attacks
- An advertiser might target very popular queries or queries with higher prior CTR
- Relevance is used to filter or re-rank ads
- Relevance(query, ad) x bid x CTR

Bids

`delicious food, news, youtube, Elon Musk, Britney Spears, …`

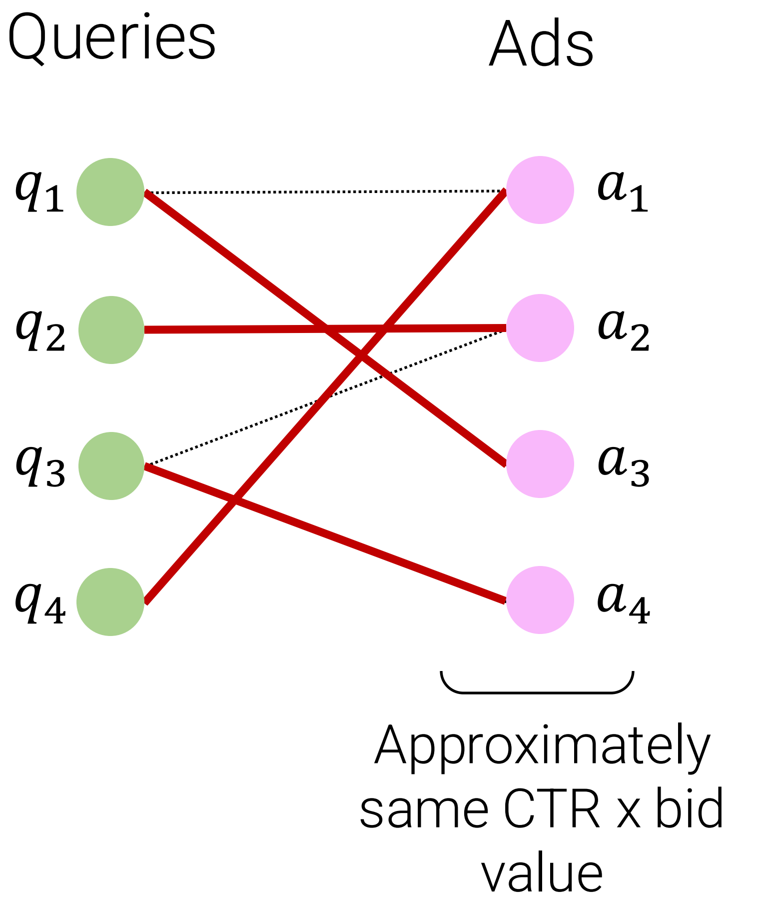


mybabyblue.com

Shady advertiser

Bipartite matching problem

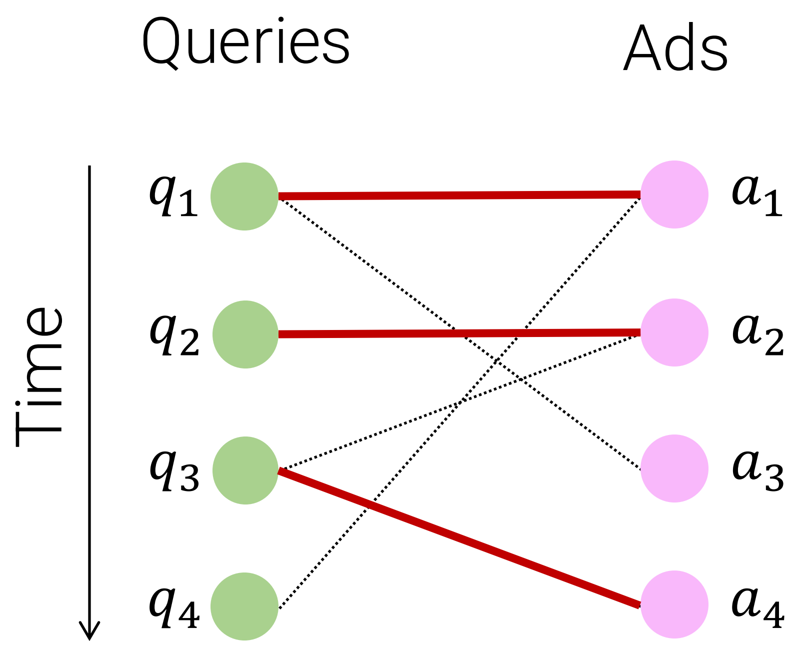
- Find a 1-to-1 match between queries and ads
- Possible matching
  - Determined by bids + relevance
- Perfect matching
  - All nodes are matched – not always possible
- Maximum matching
  - A matching that covers the maximum number of nodes given the available set of possible matchings
- When the full graph is known, an exact algorithm can be used to find the maximum matching



https://en.wikipedia.org/wiki/Hopcroft%E2%80%93Karp_algorithm

Online bipartite matching (greedy solution)

- Ads are known, query arrive in a streaming fashion
- At time t
  - A new query comes in and possible matches are calculated
  - An arbitrary query-ad pair is selected among all eligible ads
  - Example:
    - q_1 comes in, a_1, a_3 possible matchings, (q_1, a_1) selected
    - $q_2 \rightarrow (q_2, a_2)$; $q_3 \rightarrow (q_3, a_4)$; $q_4 \rightarrow none$



- Performance

  Worst-case scenario

  Cardinality of the greedy matching
  - Competitive ratio $\mathrm{CR} = \min_{i \in I} \frac{|M_{greedy}|}{|M_{optimal}|}$

  Cardinality of the maximum matching
  - Performance of greedy on the worst input compared to optimal

$CR(greedy) = 0.5$

The BALANCE algorithm

Same as greedy, but for each query, pick the advertiser with the largest unspent budget

For equal budget and equal bids:

For 2 advertisers:

$$CR(BALANCE) = \frac{3}{4} = 0.75$$

For N advertisers:

$$CR(BALANCE) = 1 - \frac{1}{e} = \boxed{0.63}$$

Best CR attainable for online algorithm

Time

Queries	Matched ad
q_1	a_2
q_1	a_1
q_1	a_2
q_1	a_1
q_2	a_2
q_2	a_2
q_2	–
q_2	–

Budget

Advertiser A
Bid \$1 on q_1 for a_1

4\$

Advertiser B
Bid \$1 on $\{q_1, q_2\}$ for a_2

4\$

The A/B testing workflow

	Activities	Examples/notes
Preliminaries	Define Driver and Guardrail Metrics	E.g., CTR vs. weekly visitors
	Design simple changes	E.g., color. Avoid multiple confounders
	Define randomization units	E.g., Users, sessions, …
Design	Define candidate set	E.g., All vs. segment (region, workflow)
	Determine sample size	How large to be confident about results?
	How long to run the experiment	Account for seasonality, day of week, …
Run	Logging	E.g., Count clicks across conditions
Results	Sanity checks	Unforeseen confounders (e.g., latency)
	Practical and statistical significance	Is effect large enough to adopt change?
Decision	Tradeoffs between metrics	E.g., higher CTR vs. lower weekly visitors
	Assess cost of the change	E.g., spending in extra compute power
	Make a recommendation	Report results and adopt, reject, or rerun
Post-launch	Measure short vs. long term	E.g., CTR lift fades away after 6 months

Metrics

- **Goal metric**: the main company mission
  - Usually not directly measurable
- **Driver metric**: the short-term objective to maximize
  - E.g., CTR, revenue, number of sales, etc.
- **Guardrail metric**: metric that should signal undesired outcomes
  - E.g., number of returning customers
- Multiple metrics measured simultaneously
  - 1 to 5 is typical
  - Sometimes combined into a single Overall Evaluation Criterion
- Metrics should be simple, clear, actionable

When different Driver metrics give opposite signals, a civil discussion between stakeholders is initiated

Statistical significance of treatment vs. control

- The two-sample *T*-test is used to verify if the population means of the treatment (T) and control (C) groups are equal or not.
- Hypothesis:
  - $\delta\text{:}\, \mu_C - \mu_T$
  - $H_\mathbf{0}\text{:}\, \delta = \mathbf{0}$. Null hypothesis
  - $H_\mathbf{1}\text{:}\, |\delta| \neq \mathbf{0}$. Alternative hypothesis
- Assumptions
  - Populations follow normal distribution
  - Populations have the same variance
  - Populations are independent
- P-value
  - Probability of observing a difference when there is none (false positive)

Statistical significance: T-test

- T-statistic:

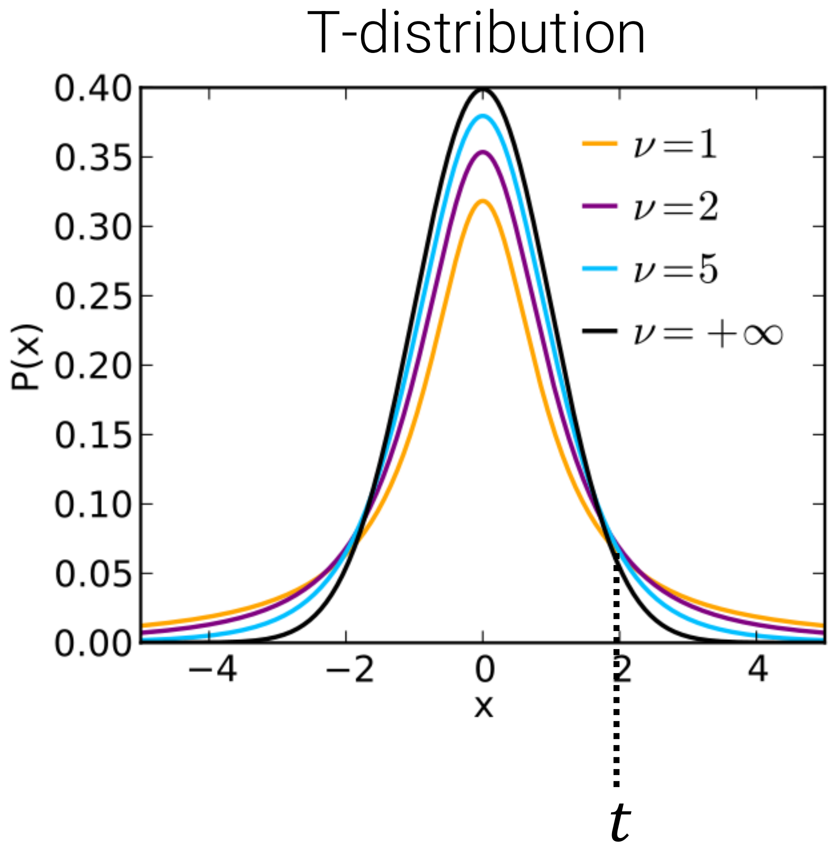
$$t = \frac{\hat{\mu}_C - \hat{\mu}_T}{\sigma \cdot \sqrt{\left(\frac{1}{n_C} + \frac{1}{n_T}\right)}}$$

Sample means of control and treatment

Pooled stdev

Number of samples in control and treatment

- P-value is $p = 2P(T_{\nu=n-1} \geq t)$
  - i.e., Two times the probability of a T-student distribution with $n-1$ degrees of freedom $(n = n_C + n_T)$ having values larger than t




```
from scipy.stats import ttest_ind
from statsmodels.stats.weightstats
     import ttest_ind
ttest_ind(x,y)
```

Statistical significance: Z-test

```
from statsmodels.stats.weightstats
     import ztest as ztest
ztest(x,y)
```

- The two-proportion *z*-test is used to verify if the proportions calculated for the treatment (T) and control (C) are equal or not.
- Hypothesis:
  - $H_0 : \pi_C = \pi_T$. Null hypothesis
  - $H_1 : \pi_C \neq \pi_T$. Alternative hypothesis
- Assumptions
  - Populations follow normal distribution
  - Populations are independent
- P-value
  - Probability of observing a difference when there is none (false positive)

Check against normal distribution to get p-value

Proportions of control and treatment

$$z = \frac{\hat{\pi}_C - \hat{\pi}_T}{\sqrt{\hat{\pi}(1 - \hat{\pi}) \cdot \left(\frac{1}{n_C} + \frac{1}{n_T}\right)}}$$

Proportion of pooled sample

Workflow

1. State expected difference δ that you would like to observe
   - δ can be decided by stakeholders (e.g., interested in a minimum of 2$ lift)
   - δ can be informed by A/A test (e.g., observed difference of 0.5$ in A/A)
2. Estimate standard deviation of your population from historical data
3. Run a power analysis to learn the minimum sample sizes for control and treatment
4. Run the A/B test and collect results
5. Run the test longer (and for more samples) than the minimum to ensure consistency and sufficient representativity
6. Measure statistical significance of the difference with a T-test
7. Measure the practical significance of the difference
8. Reach a verdict on the value of the new solution
9. If verdict is positive, gradually ramp up the treatment to larger portions of the user base

Common A/B testing mistakes

- P-value peeking
  - Run the A/B test and stop it as soon as p-value is < 0.05: **WRONG!**
    - Results can fluctuate and appear significant earlier than the minimum sample size is collected, but they can drop again later
  - Instead: run experiment as long as designed (or more)
- Multiple testing problem
  - When testing multiple driver metrics
  - Solutions:
    - Bonferroni correction
    - Tiered significance (assign lower p-values to important metrics)
- Violating Stable Unit Treatment Value Assumption
  - Randomization units must be independent, and have no interaction
  - Easily violated in social networks or in two-sided markets (when change in demand/supply in treatment may affect control)

Bandits vs. ab test

- Use A/B testing to **understand** the effect of a treatment
  - Long-lasting design choices (a new interface, a new module)
- Use Bandits to **optimize** among many, quickly varying alternatives
  - Optimization of an ad or a headline (need decision quickly, as content will expire)
  - Recommendation systems

- Bandits spend less time picking sub-optimal choices than A/B testing, but they take longer to get statistical significance for all possible choices

Contextual Bandits exercise

- See how a LinearUCB bandit can be trained using a python library
- Rewrite the method to run the replay evaluation for contextual bandits
- Pandas <2.0 might be needed for a successful run

A/B testing exercise

- Test a simulated A/B scenario on revenue lift, try to see how results (sample size, significance) change when changing parameters
- Replicate (almost) the same workflow for a scenario of CTR lift